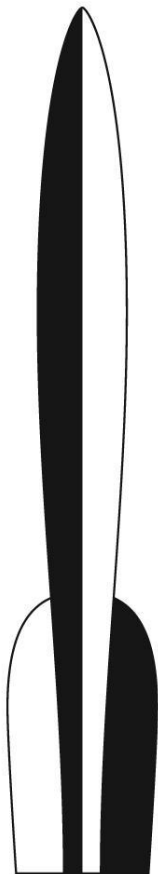


What is Nomination Worthy?



Yilda Boukhtouchen
Annika Christiansen
Xingyu Cheng
Jay Desai
Qisi Zhang

Predicting the success of newly-released fantasy & sci-fi novels



Introduction

Aim: To predict if a given new science-fiction and fantasy (SFF) book release will be ***nominated*** as a finalist for the

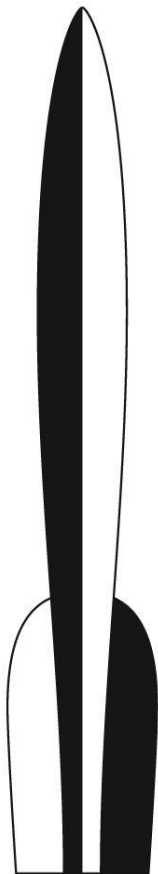
- **Hugo Award for Best Novel**
- **Locus Awards for Best Fantasy & Best SF Novel**

Stakeholders:

- Publishers & booksellers
- Authors and readers

KPIs:

- Primary: F_1 score
- Secondary: precision, NDCG for ranking models



Data Collection & Cleaning

**International Speculative
Fiction Database (ISFDB)**
SQL data dump of book metadata

Hardcover App API
genre tags and book descriptions

**Science Fiction Award
Database (SFADB)**
yearly published top nominees

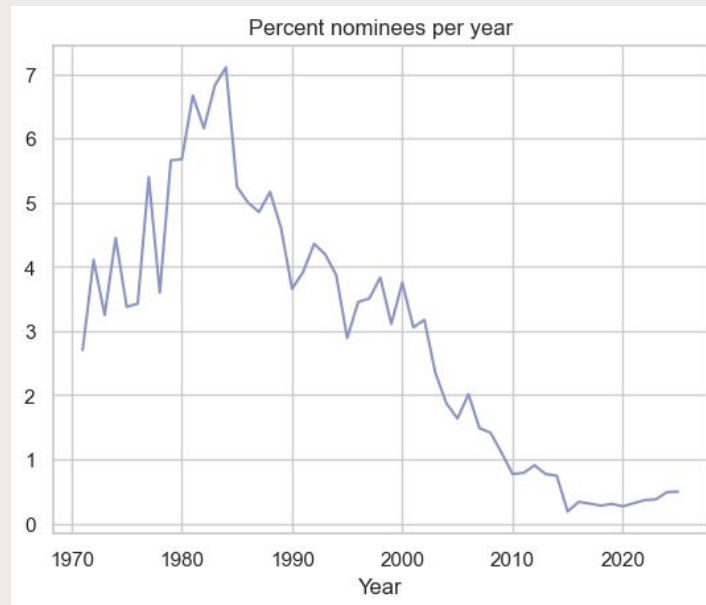
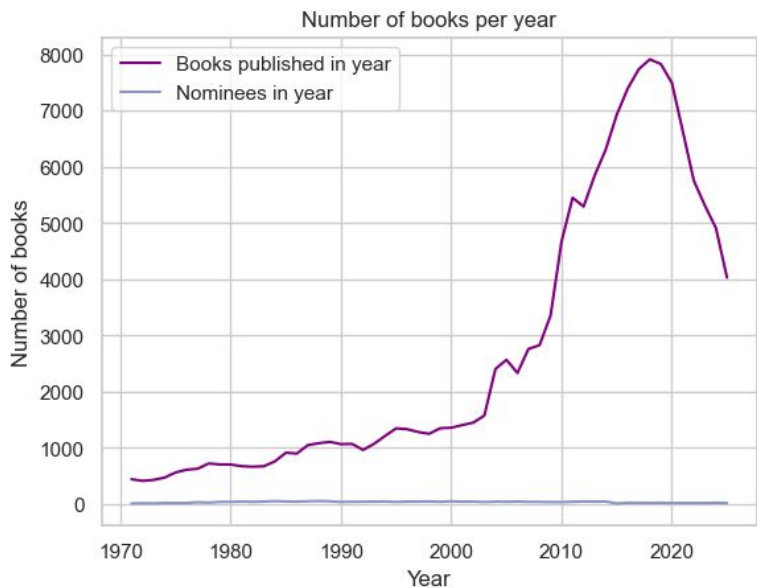
Data Merging & Cleaning
find first published editions
check against award eligibility criteria
resolve title mismatches between databases
remove data leakage from tags
clean up tags
years: 1971 (Locus inception) to 2025

Training Set
1971 - 2014

Test Set
2015 - 2025

Data Exploration

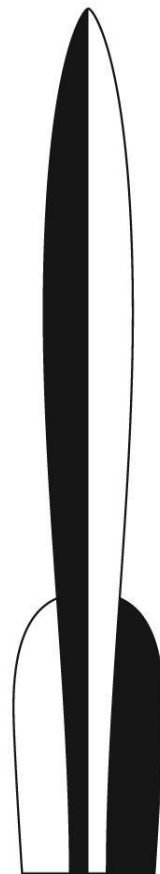
We have a class imbalance (i.e. need to be careful of models and metrics to use)



Data Exploration: Engineered Features

	Features	Processing
Author	previously nominated (bool), past nomination count, past novel count	z-score normalization compared to year cohort
Publisher prestige	past nomination count, cohort size (# books published that year)	z-score normalization compared to year cohort
Publication details	First publisher, publication month	one-hot encoding
Genre similarity	Cohort similarity, past-nominee similarity	cosine similarity in tag vector space

To avoid time leakage, all features only normalized to books within same year.

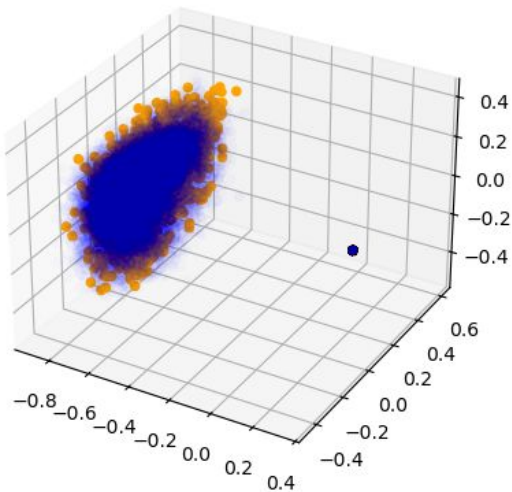


Data Exploration: Engineered Features

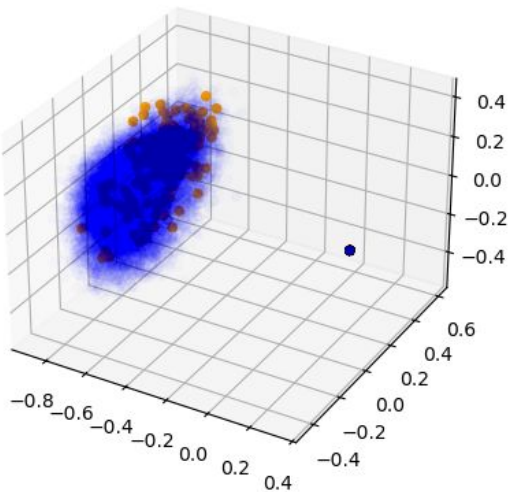
For book descriptions, we used a pre-trained transformer [all-MiniLM-L6-v2](#) to encode book descriptions into a 384 dimensional vector space.

Using PCA to visualize 3-dimensional slice of the data:

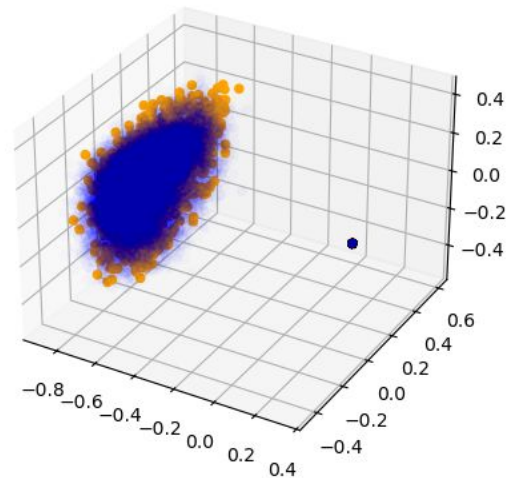
Locus vs non-Locus

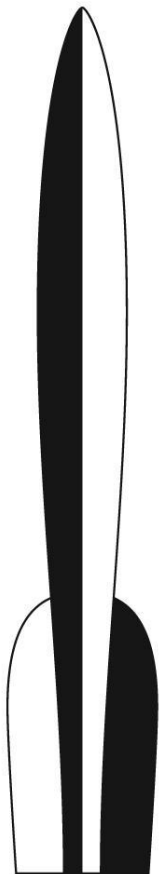


Hugo vs non-Hugo



Nominees vs non-nominees





Baseline Models

DummyClassifier with stratified strategy.

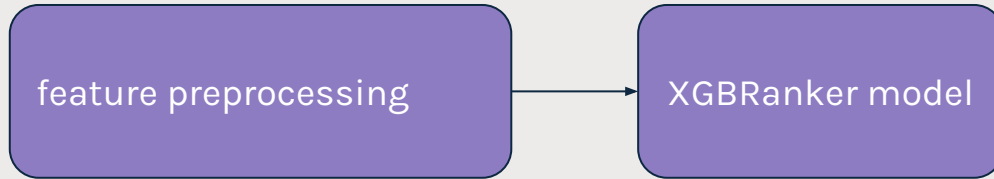
Naïve prior-nomination baseline: predict that only novels written by a previously-nominated author will be nominated again.

We also consider a standard **logistic regression** model.

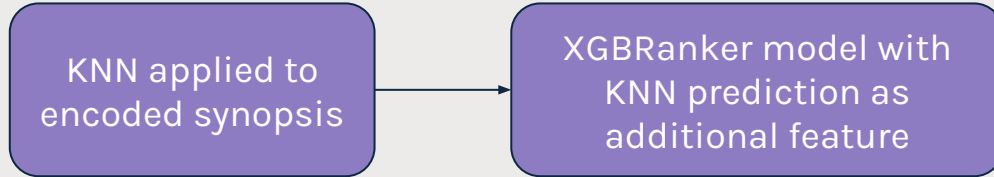
We employ **walk-forward nested cross-validation**.

Final models

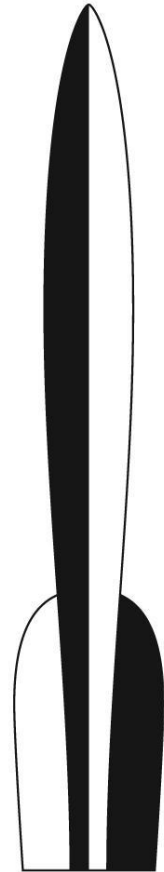
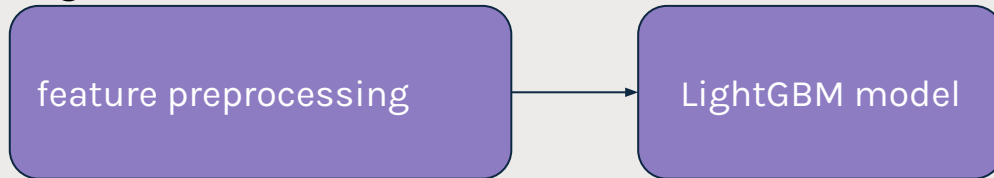
XGBRanker model



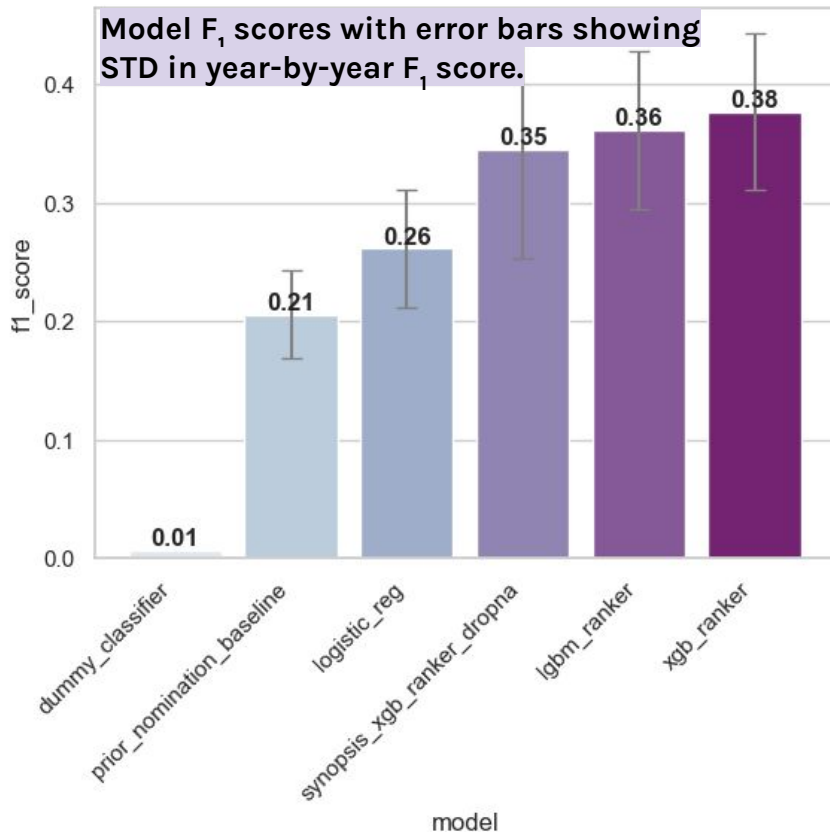
SynopsisXGBRanker model



LightGBM model

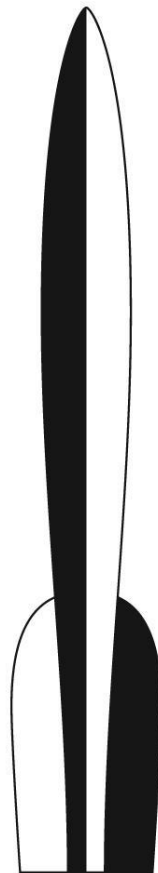


Model Evaluation Results

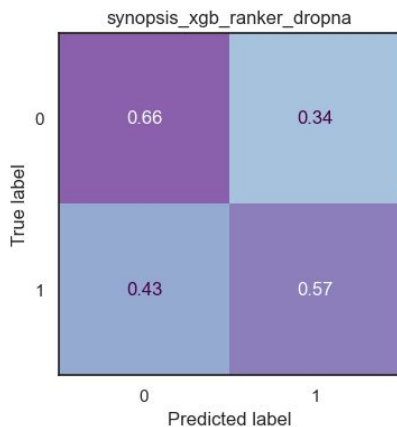
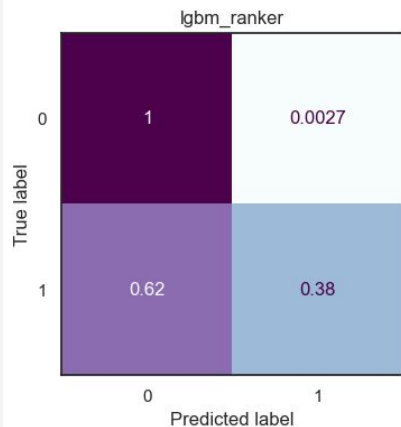
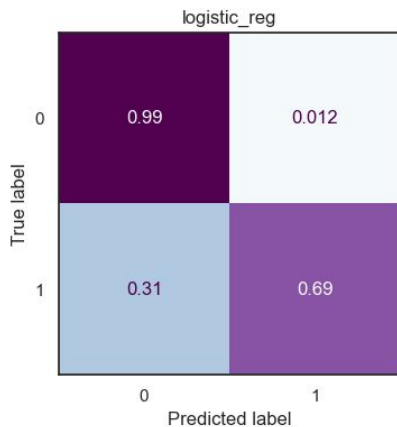
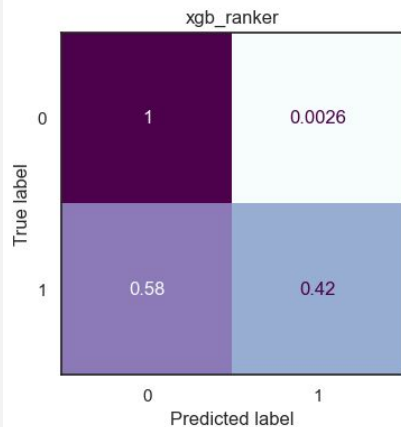


LearnToRank models generally outperform baseline models.

SynopsisXGBRanker performs well only **after** dropping missing fields.



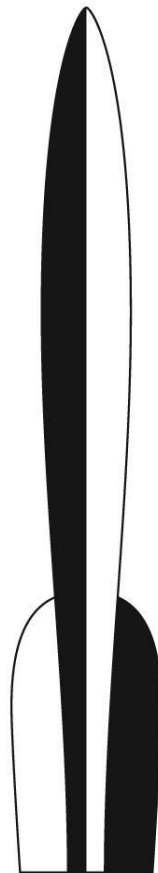
Model Evaluation Results



LearnToRank models on full test set have **lowest false positive count.**

~40% precision rate in top 10 predictions.

SynopsisXGBRanker has **higher true positive rate**, balanced by more false positives.



Model Limitations & Future Directions

Difficulty with new author nominees	Early popularity metrics
Author nomination metrics are highly weighted in many of the models.	ARC reviews, early ratings on Goodreads or similar, pre-release media articles.
Year-by-year variability & data drift	Book-related social media trends (or fan-magazine trends in pre-90s)
Trends within the genre, publishing landscape and fan engagement methods vary over the years.	Allow further quantification of cohort similarity/uniqueness, and context on popular subgenres/authors.